

Electronic Inspirations: Technologies of the Cold War Musical Avant-Garde

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CHAPTER

2 Kinship: Cage, Tudor, and the New Timbral Utopia

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Abstract

WDR composers drew from several sources in creating their electronic timbral utopia, including John Cage's early works. His compelling prepared piano pieces provided a model for the new sounds of early electronic music by Schaeffer, Stockhausen, and Eimert. Furthermore, Cage's square-root form provided a model for Stockhausen's and Koenig's handling of duration in their early tape music, especially because Cage embraced a proportional approach to pitch–time relationships. Although Cage and Tudor's 1954 European tour to Donaueschingen and Cologne was partly a scandal, the pair were valued as celebrities by power brokers in the European milieu. Cage became more of a provocateur in the later 1950s, growing distant from the younger avant-garde composers. Tudor, however, remained exceptionally important as a nodal figure in the new music network, whose prodigious and consistent performances knit together the American and European avant-garde scenes in the 1950s and 1960s.

Keywords: John Cage, David Tudor, prepared piano, square-root form, scandal, celebrity, reception, new music networks

Subject: Musicology and Music History

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The oft-repeated story about the American experimental composer John Cage and the 1950s European avant-garde (led by Stockhausen and Boulez) goes something like this: Cage and the Europeans repelled each other like oil and water because Cage's interest in chance and indeterminacy was incompatible with the dogmatic European serialists, who were striving for total control of musical materials.¹ This familiar narrative of rivalry uses as its primary point of evidence John Cage's incendiary visit to the Darmstadt *Ferienkurse* in September 1958. Cage and pianist David Tudor (his “invisible” collaborator) presented a concert of two-piano works that was met with a mixture of genuine appreciation and heckling.² Cage then delivered three lectures, ostensibly on his own indeterminate compositional practices and philosophies, but peppered throughout with barbs against the compositional practices and attitudes of Stockhausen, Boulez, and others.³ His critiques were made only more provocative by hasty, inelegant translations, which amplified their confrontational tone.⁴ As

Christopher Shultis writes, “No other event in Darmstadt’s history ever generated more controversy than Cage’s 1958 lectures.”⁵ This narrative places Darmstadt as the epicenter of the European avant-garde and concludes that the Europeans received Cage’s work somewhere between ambivalence and antagonism.

In this chapter, I encourage an expanded focus—temporally, geographically, and interpersonally—in order to rethink the relationship among Cage, Tudor, and the European serialists. Although Darmstadt 1958 was a visible controversy, it was not Cage’s first or most important encounter with the European avant-garde. Between 1949 and 1954, Cage enjoyed a warm courtship, lasting relationships, and significant celebrity among a small but powerful segment of the European avant-garde. This early embrace of Cage was especially apparent in the emerging WDR studio, where Cage’s prepared-piano works were crucial to the development of electronic music. (A prepared piano is a piano in whose strings various objects or “preparations” like screws, bolts, and rubber have been placed). Moreover, after Cage’s relationships with some of his European colleagues became more strained in the late 1950s, Tudor remained exceptionally important as a nodal figure, connecting American and European strands of the avant-garde well into the 1960s. Cage and Tudor’s contributions, especially in the milieu of the WDR studio, were important facets of the emerging discourse of electronic music.

Past Tense

Cage’s first encounter with the European avant-garde was in Paris in June and July 1949, when Cage and Merce Cunningham presented *Sonatas and Interludes* and *Three Dances* for prepared piano, among other works, in Olivier Messiaen’s class, in Jean Heli  n’s salon, in Suzanne T  zenas’s salon, and at the Th   tre du Vieux-Columbier.⁶ Boulez, a close associate of the socialite and patron T  zenas, gave a spoken introduction to Cage’s work at her salon; afterward, Cage and Boulez maintained a lively correspondence and a close friendship for several years. In 1951, when the two were exchanging warm letters frequently, their compositional techniques overlapped closely. Both composers used gameboards or other precompositional charts in part to minimize their own moment-to-moment choices. As Boulez wrote to Cage, “I’ve taken over your chess board system for my own purposes, by making it serve on dissociated, antagonistic, and parallel or anti-parallel levels.”⁷ Cage replied in solidarity, “By making moves on the charts I freed myself from what I thought to be freedom, and which actually was only the accretion of habits and tastes.”⁸

As this exchange shows, Boulez and Cage shared a desire to move beyond their own inherited instincts. This is a well-understood aspect of Cage’s poetics.⁹ In his own words, throwing the dice allowed him “to make a composition the continuity of which is free of individual taste and memory (psychology) and also of the literature and ‘traditions’ of the art.”¹⁰ Achieving such aesthetic freedom was also a major preoccupation of the postwar European avant-garde. As Boulez said, “dodecaphonists and independents alike work to inculcate liberty. [. . .] Any musician who has not experienced—I do not say understood, but truly experienced—the necessity of dodecaphonic language is USELESS.”¹¹ When Boulez and Cage argued the “necessity” of compositional systems based on their affordances of liberty and freedom, they revealed that their aesthetic choices were implicitly linked to historically conditioned, political–aesthetic circumstances.

In short, the appeal of such systems, whether serial or aleatoric, lay in their ability to neutralize the trained, conservative, or (morally) corrupt instincts of composers. The younger generation of avant-garde composers were born in the 1920s and were reared and educated during the Nazi era, when racial exclusion, nineteenth-century aesthetics and tonality, and electronic music instruments were packaged together into a grotesque mandate. In Germany, those in the older generation had to suppress their new music interests (as Eimert did), win the favor of the Nazi Party (as Trautwein did), or leave the country (as Schoenberg did). This bind was recast in the cultural Cold War, as the Soviet Socialist Realist doctrine again mandated conservative artistic positions and avant-garde art symbolized democratic freedom.¹² As Richard Taruskin summarizes, in the 1950s avant-gardism “seemed to embody a perfect artistic ‘autonomy.’ That autonomy easily translated into

personal and political autonomy—that is, individual integrity—in the minds of many who were emerging from decades of oppression, an oppression that was still going on in the East.”¹³ Cage and Boulez began as kindred spirits, who in the early cultural Cold War mentality, used rigid systems to unlock and perpetuate their liberty—a paradox that reveals its own historically contingent logic.

The fellow travelers did not remain so for long. The first cracks appeared in December 1951, when Boulez wrote to Cage, “The only thing, forgive me, which I am not happy with, is the method of absolute chance (by tossing the coins). [. . .] I am a little afraid of what is called ‘automatic writing’ for most of the time it is chiefly a lack of control.”¹⁴ Boulez further registered his discontent along similar lines in 1954.¹⁵ With the publication of “Alea” in 1957 Boulez publicly amplified his complaints and set the stage for the Darmstadt debacle.¹⁶ However, to focus only on the progression of the aleatory debates, extending outward from the personal relationship between Cage and Boulez, obscures the degree to which Cage was a key figure to many of the European composers between 1949 and 1954. In this era, aided by Boulez’s introductions, Cage and Tudor gained significant traction with Eimert, Stockhausen, and many other powerful impresarios and composers in the West German scene.

Widening Circles

Figure 2.1 shows a timeline of Cage’s exposure by means of West German new music institutions between 1952 and 1954. Cage and Tudor garnered cachet in West Germany through Eimert’s WDR *Musikalisches Nachtprogramm* and subsequently through other radio broadcasts (Otto Zoff on the SWF [Südwestfunk, or Southwest German Radio]) and lectures (Wolfgang Rebner at Darmstadt).¹⁷ Radio was an extraordinarily important medium at that time, and it would be difficult to overstate the impact of Eimert’s particular endorsement of Cage on his *Nachtprogramm* in November 1952. Eimert’s twice-monthly broadcasts, which aired from 11:15 p.m. to midnight and attracted a wide range of listeners, were critical in disseminating new music that had been forbidden or forgotten, as well as analysis of it. The show was something like a correspondence course for keeping current with new music; as Martin Iddon says, “In short, anyone who was anyone was listening.”¹⁸ Consequently, using the *Nachtprogramm* as something of a bully pulpit, Eimert had substantial power to shape the narrative and direction of the postwar musical avant-garde. This West German interest in Cage culminated in a pair of concerts in fall 1954 at Donaueschingen’s *Musiktage* and Cologne’s *Musik der Zeit*, where (at the latter) Cage and Tudor’s stunning prepared-piano pieces were heard opposite the newest electronic music from the WDR studio.



Figure 2.1. Timeline of Cage and Tudor’s West German exposure, 1952–1954.

In his 1952 broadcast, Eimert set the stage by introducing Cage to West Germany with a reasonably clear (though somewhat stereotypical) understanding of his aesthetic and philosophical premises, including mentions of gamelan, Meister Eckhardt, Buddhism, and silence.¹⁹ Eimert portrayed Cage as a descendant of jazz music, Buddhism, and a lineage of American experimentalists such as Edgard Varèse, Henry Cowell, George Antheil, and Arnold Schoenberg.²⁰ Although Schoenberg was an Austrian who emigrated to the United

States and gained citizenship in 1941, and although Antheil was immersed in the Parisian avant-garde in the 1920s, both played into Eimert's narrative of American rugged individualism and aesthetic freedom. As he wrote to Stockhausen contemporaneously, "Cage's works are interesting and completely unencumbered by conventional concepts."²¹ According to the Europeans, Cage admirably stood outside of the musical tradition that they were all trying desperately to navigate and sidestep.

p. 53 As Table 2.1 shows, most of the musical examples broadcast on Eimert's program were piano works—especially two-piano and prepared-piano works. Eimert contextualized these as examples of an "American experimental noise" heritage that extended from Varèse through to its fullest elaboration in Cage's *First Construction*; he concluded the broadcast with an excerpt from Bruno Maderna's two-piano concerto. Eimert probably focused on two-piano works because Boulez gave him a record in 1952 that must have contained Cage's *Three Dances*, *A Book of Music*, or both—a record of prepared-piano works for two pianos.²² Boulez had already spoken about these pieces, as well as Cage's *Sonatas and Interludes*, in his introduction to Cage's 1949 performance at Tézenas's salon, so it stands to reason that Boulez had recordings of them in his possession.²³ Eimert also played from what he claimed to be a Cage work for two pianos tuned a quarter-tone apart. No such piece by Cage is known, and it seems likely that what Eimert mistakenly attributed to Cage—a work that perhaps also appeared on Boulez's record—was actually Charles Ives's *Three Quarter-Tone Pieces for Two Pianos*.²⁴

Table 2.1. Repertoire played during Eimert's November 27, 1952 *Musikalisches Nachtprogramm* broadcast.

Musical Example and Composer	Piece as Named in Broadcast Transcript	Corresponding Work
1: Cage	"kleines Beispiel aus Cages Stücken für präpariertes Klavier"	<i>Three Dances</i> (for two amplified prepared pianos) (1944–1945)? <i>A Book of Music</i> (for two prepared pianos) (1944)? <i>Sonatas and Interludes</i> (1946–1948)?
2: Cage	"Festmusik für prepared Piano"	<i>Bacchanale</i> (1938–1940)
3: Varèse	<i>Ionisation</i>	<i>Ionisation</i> (1929–1931)
4: Cage	Konstruktion in Metall	<i>First Construction (in Metal)</i> (1939)
5: Cage	Konstruktion in Metall, 2nd example	<i>First Construction (in Metal)</i> (1939)
6: Cage [Ives]	"kleine Vierteltonpassage"	Ives, <i>Three Quarter-Tone Pieces for Two Pianos</i> (1923–1924)
7: Cage [Ives]	"ein Beispiel aus John Cages Vierteltonmusik für zwei Klaviere"	Ives, <i>Three Quarter-Tone Pieces for Two Pianos</i> (1923–1924)
8: Maderna	Konzert für zwei Klaviere und Instrumente	Concerto (for two pianos and instruments) (1948)

p. 54 Eimert characterized the prepared piano as the most recent stage in a long evolution of keyboard instruments, using the same type of historicizing narrative meant to justify electronic music and other new sounds to the "stubbornly conditioned ears of the public," in Varèse's famous phrase.²⁵ Eimert viewed Cage as a brother in arms in the quest to transcend the limitations of acoustic instruments, and yet, he did not hold Cage's electronic work in high esteem. As Eimert wrote rather pejoratively to Stockhausen, "His [Cage's] electronic tapes [*Imaginary Landscape No. 5* (1952) or *Williams Mix* (1952–53)?] are by contrast quite primitive and contain

nothing more than a few feedback tones.”²⁶ Nevertheless, Eimert found a conceptual and even spiritual kinship between Cage and European composers such as Boulez and Messiaen: all were searching for an “Authentic Music,”²⁷ a “music structured according to a crystalline logic without psychology, without memory.”²⁸ In Eimert’s reading, Boulez and the European avant-gardists sought this aim by magnifying Webern’s pointillist aesthetics with integral serialism, whereas Cage and the American experimentalists pursued it by allowing the structure of the piece to spring from the interplay between new sonic materials, like the prepared piano and silence.

Preparing Electronic Music

At the time of Eimert’s Cage broadcast, the WDR electronic studio space was anxiously anticipated. “What a shame that the construction of the WDR electronic studio is proceeding with such bureaucratic slowness,” Eimert complained to Stockhausen in March 1952, “it is still months to completion.”²⁹ Although the studio had been in the planning and construction phases since 1951, the first pieces would not premiere there until May 1953. While Eimert waited for the studio to become a fully functional space, he and Beyer were busy exploring the new timbres of the Melochord and montaging the sound catalogues of Meyer-Eppler and Schütz into their first pieces.

Meanwhile, Eimert wrote frequently to his protégé Stockhausen, who was, at the time, working as Pierre Schaeffer’s intern at *Radiodiffusion Française* in Paris. Stockhausen was Eimert’s main informant for contemporary French studio developments—an important role because Eimert was continually evaluating how much ground he was losing to the French, who had been making electronic music since the late 1940s. Stockhausen fielded anxious questions such as, “What is Schaeffer up to now? What about Messiaen?”³⁰ “It is important to me to learn about Boulez’s electronic experiments,”³¹ and “Write to me more about Boulez, please.”³² The West Germans anticipated their own electronic studio at the WDR with ebullient excitement but also plenty of competitive hand-wringing, primarily over their self-perceived lateness. It was in this environment that Cage’s prepared-piano works were important models for the *terra incognita* that was early electronic music.

Eimert’s combination of anticipation and anxiety was shared by many of those active in electronic music circles, whether in West Germany, France, or the States. In this era, composers and technicians confronted unfamiliar and unwieldy machines holding fast, on the one hand, to the expansive promises of new sounds while struggling, on the other hand, to find palatable ways to produce electronic music. In his own early experiments, Pierre Schaeffer searched for appropriate sound sources and sonic models for *musique concrète*. As Schaeffer wrote in his journal in June 1948,

There is no instrument on which to play concrete music. That is the main difficulty. Or else we have to imagine a huge cybernetic-like machine that can achieve millions of combinations, and we’re not there yet. As long as I have no more than two or four turntables that make only approximate transitions, I shall remain horribly imprisoned in a discontinuous style where everything seems to have been hacked out with a billhook. Is there a compromise? I instinctively turn to the piano. The preceding manipulations have in fact taught me that a piano could well replace all sound effects equipment. You can strike the strings directly, or scratch or lightly stroke them but you can also use the keyboard, not as a musical instrument, but as a convenient way of attacking the strings, which will have undergone some “preparation.” [Schaeffer’s footnote: the term “prepared piano” was used systematically by the American John Cage, whose works I was unaware of at this time. The use of the piano is similar in both cases, but with John Cage it leads to a music that still remains fairly abstract in its conception and performance].³³

Dissatisfied with the denaturing effects that could be induced with the studio's turntables, Schaeffer imagined the prepared piano as an already complex, already noisy instrument that would serve as a super-percussion section and a sophisticated sound source for his *musique concrète* works. The footnote, apparently inserted retrospectively when Schaeffer prepared his journals for publication in 1952, suggests that he hadn't yet encountered either the concept or the terminology of Cage's prepared piano when he wrote this journal entry in 1948. We can surmise, however, that Schaeffer did hear Cage's prepared piano in one of the four Paris concerts in June 1949, where attending avant-garde composers called the sounds "profound." As Karel Goeyvaerts remembered, "The crisp sounds of his gamelan piano and the precise rhythm of the sonatas kept us spellbound. Messiaen claimed that this was his most riveting musical experience since he first discovered Çarngadeva's *Deçitâla*."³⁴

Stockhausen, who must have encountered Cage's prepared piano on a recording in Schaeffer's studio in 1952, began composing his own preliminary *musique concrète* experiment in close quarters with Goeyvaerts and using the prepared piano as his sound source.³⁵ The experimental piece may have "ended in catastrophe,"³⁶ but the point is that Stockhausen, like Schaeffer, considered the prepared piano a promising sound source for electronic music. Boulez, too, seemed taken with the new sound world that Cage's prepared piano had opened up:

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His use of the *prepared piano* is not merely an unusual area of percussive piano technique in which the sounding board is to be invaded by a strange, metallicizing vegetation. It has much more to do with calling into question acoustic ideas which have gradually become fixed in the course of Western musical evolution; for this prepared piano becomes, by way of a do-it-yourself tuning system, an instrument capable of producing frequency complexes.³⁷

Boulez gets to the heart of the matter: the prepared piano was beguiling because its sounds unsettled the assumptions and expectations of listeners. The prepared piano married familiar old gestures to intriguingly new sounds. It displaced the predictable timbres of the acoustic piano, while remaining in a dialectical relationship with the historical progression of instrument development. This, of course, was the same dream that composers held for electronic music. In Boulez's formulation and in Schaeffer's (previously quoted), the prepared piano can almost substitute for the studio; both scenarios were capable of producing "frequency complexes." As the repertoire on the *Musik der Zeit* concert of 1954 suggests, it was only one small step from the new timbres of Cage's prepared piano to electronic music (see Figure 2.2).



Figure 2.2. Program for *Musik der Zeit*, October 19, 1954. Public premiere of the second group of electronic pieces from the WDR studios. WDR Historisches Archiv.

The aural similarity between early WDR electronic music and Cage's prepared-piano music of the same era is compelling. Consider Eimert's *Glockenspiel* (1953), the eighth piece to be produced in the WDR studio, and the first piece for which Eimert is credited as the sole composer, without Robert Beyer. *Glockenspiel* shares a number of conspicuous sonorities with Cage's prepared-piano timbres, which is perhaps not surprising, as it was likely composed only a few short months after Eimert's *Nachtprogramm* broadcast. We might use Cage's *Bacchanale* (1938–1940) for comparison, because Eimert played it during the broadcast and could have had the piece in mind (and on record) as he composed *Glockenspiel*.

There are at least three sound-colors that are strongly shared between *Glockenspiel* (German for bells, chimes, or carillon) and *Bacchanale*: muted sounds, bell sounds, and metallic sounds. The muted sounds, produced when fibrous weather stripping is placed between the strings to dampen the vibrations, is particularly pervasive in *Bacchanale*, used in ten out of twelve preparations.³⁸ In *Glockenspiel*, the muted timbre is also pervasive, characterizing many of the low-frequency, gong-like strokes that begin the piece and are interspersed throughout. The many bell-like and metallic sounds shared between the pieces are produced on the prepared piano by inserting screws or bolts in the middle to low register, such as the third octave; one preparation (on D4) involves a “small bolt” whereas another (on Bb3) involves a screw with nuts plus weather

p. 57 stripping.³⁹ In the electronic studio, on the other hand, the bell-like and metallic timbres are almost surely a result of ring modulation, in which two frequencies are added together. ↵

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The input frequencies remain unheard under ring modulation, and the heard output is the sum and difference of the input frequencies, which creates an inharmonic relationship and thus a bell-like timbre.⁴⁰

It is unlikely that Eimert arrived at these inharmonic, bell-like sounds through additive synthesis.⁴¹ As Chapter 3 will show, it was not until the end of 1953 and early 1954 that the younger WDR composers, such as Stockhausen, Pousseur, Gredinger, and Goeyvaerts, collaboratively stumbled toward a workable solution for synthesizing inharmonic mixtures. We can speculate, based on the resources of the WDR studio and our knowledge of Eimert's compositional comfort zone, that the sounds for *Glockenspiel* were produced on the Melochord, which contained a number of generators and filters activated through knobs and a two-manual keyboard (refer back to Figure 1.1). The Melochord's bottom manual produced tones from a tone generator, whereas the top manual modified the sound. These modifications could include filtering (such as high-pass, low-pass, and bandpass); ring modulation; certain sound envelope characteristics such as plucked or blown attacks; and vibrato, noise, and reverberation. The various dynamic filters of the Melochord could certainly account for the muffled, muted timbres in *Glockenspiel*, and the instrument's ring modulator could have been used to produce metallic and bell-like tones. In summary, the timbral vocabulary of Cage's *Bacchanale*, Schaeffer's early *musique concrète*, and Eimert's *Glockenspiel* are remarkably similar. All of this gives aural evidence that Cage's idiosyncratic prepared-piano sounds provided an aesthetic model for Eimert—as well as for Schaeffer, Boulez, and Stockhausen—in their earliest attempts to create a new, electronic timbral world.

Cage as Celebrity

The initial exposures that Cage garnered in Europe in the late 1940s and early 1950s—in France by means of Boulez and Schaeffer on the one hand and in West Germany by means of Eimert and radio broadcasts on the other—seem to have accrued to substantial prestige by October 1954. Cage and Tudor were invited to play their prepared-piano works at Donaueschingen by Heinrich Strobel, “arguably the most powerful administrator in West Germany's new music community at the time.”⁴² Strobel was appointed music director of the SWF Baden-Baden in 1945, served as editor for the popular new music journal *Melos*, and organized the aforementioned long-standing prewar and postwar new music festival at Donaueschingen.⁴³ After receiving Strobel's invitation, Cage and Tudor, wanting to “stir up as much activity as possible” while in Germany, reached out to Eimert, expressing interest in also visiting the electronic music studio and arranging a “radio engagement” at the WDR.⁴⁴

p. 59 Cage ventured into rather fraught waters in trying to arrange events in both Donaueschingen and Cologne. Strobel had initially offered Cage a forty-five-minute concert, and Cage, rather presumptuously, countered with a proposal for *three* programs, one for piano, one for orchestra, and one for tape. Strobel stuck to his initial offer for one program. The orchestral aspect of Cage's proposal was abandoned entirely, and works for prepared pianos and magnetic tape by Cage, Morton Feldman, Earle Brown, and Christian Wolff were programmed together on a matinee concert (see Table 2.2).⁴⁵ In Cologne, Eimert's assistant Eigel Kruttge initially offered to include Cage and Tudor in an October 8 chamber concert, but Cage could not accept because he had promised Strobel that he would play nowhere in Germany before the Donaueschingen engagement on October 17. As Strobel explained,

The attraction of your coming out in Germany, in presence of our large international auditorium would be lost, if you play anywhere in Germany before. I hope you understand that your

Donaueschingen performance only can be realized if you give no performance before our concert, neither over another radio station [n]or anywhere else in Germany.⁴⁶

Table 2.2. Program for Donaueschingen *Musiktage* matinee performance, October 17, 1954. Asterisked pieces were billed as premieres.

Composer	Title	Performer(s)
Christian Wolff	<i>For Pianos II</i>	David Tudor
Earle Brown	<i>Four Systems</i>	David Tudor
John Cage	*12'55.6078" for two prepared pianos	David Tudor and John Cage
Morton Feldman	* <i>Intersection</i>	tape
Earle Brown	<i>Octet</i>	Tape
John Cage	<i>Williams Mix</i>	Tape

These terms make clear what was at stake: Cage had considerable celebrity cachet, which Strobel was determined to draw to Donaueschingen (and away from Paris and Cologne). In courting Cage to appear at *Musik der Zeit*, Eimert (through Kruttge) tried to draw that prestige toward Cologne in an attempt to validate the WDR as the most important German (and international) new music center.

Although October 8 was out for an engagement in Cologne, Cage wrote again to Kruttge at the WDR, requesting a change of date or a change of planned program:

Since I have not had a reply from you to my last letter, I am writing to say that I hope matters may be arranged in such a way that the concert plans you outlined may be realized. For that reason, please consider our needs as stated in my response to you flexible. I have asked Dr. Strobel whether he would be willing for us to appear in Köln [Cologne] before the 17th of October providing the performance does not include a work for prepared piano. If he agrees to this, you might be interested in a performance by David Tudor of my *Music of Changes* (about 42 minutes). [. . .] And, finally, if the honorarium I requested was too high, please let me know what you will offer. In any event, be assured that I am sincerely interested in cooperation with you.⁴⁷

Having still no reply a month later, Cage wrote a third time. His anxiety at having overreached in these negotiations was palpable:

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Although I have not heard from you in reply to my last two letters, I still hope that a concert such as you outlined may take place in Köln with our cooperation. Dr. Strobel is now interested in presenting my work for magnetic tape on the Donaueschingen program. But I would prefer to have it performed at Köln and will delay answering him, hoping for an early reply from you. My present situation is that described in my second letter to you: that is: that I would like to know from you what you will offer in the way of facilities and fee, rather than that the requests of my first letter be considered inflexible (with the exception of date [. . .]).⁴⁸

Although Cage worried that he misjudged his requests for scheduling accommodations and a large honorarium, he also seemed to recognize his work was a valuable commodity. Cage clearly did not wish to appear as the *enfant terrible*, though his cooperative spirit was undermined as he tried to play the SWF off the WDR.

In the end, in a letter dated June 1, Kruttge acquiesced to Cage's requests for a change of date, and the Cologne evening concert took place on October 19, two days after the Donaueschingen matinee.⁴⁹ As Tables 2.2 and 2.3 show, the programs given at Donaueschingen and Cologne were remarkably similar. Cage and Tudor played the piano works of Cage (probably shortened versions of 34'46.776" and 31'57.9864") and several American collaborators in the first half, counterpointed by electronic music—by Feldman, Brown, and Cage in Donaueschingen and by the WDR studio composers in Cologne—in the second half.⁵⁰ Despite the similarity of repertoire, both festivals claimed they were offering premieres, with all the preeminence and symbolic capital contained therein.

Table 2.3. Program for Cologne *Musik der Zeit* performance, October 19, 1954. Asterisked piece is billed as European first performance.

Composer	Title	Performer
Morton Feldman	<i>Intersection 3</i>	David Tudor
Christian Wolff	<i>For Prepared Piano: vier kleine Stücke</i>	David Tudor
Earle Brown	<i>Perspectives</i>	David Tudor
John Cage	*23'56.176" for two prepared pianos	David Tudor and John Cage
Karlheinz Stockhausen	<i>Studie II</i>	Tape
Herbert Eimert	<i>Glockenspiel</i>	Tape
Karel Goeyvaerts	<i>Komposition Nr. 5</i>	Tape
Henri Pousseur	<i>Seismogrammes I</i>	Tape
Paul Gredinger	<i>Formanten I und II</i>	Tape
Karlheinz Stockhausen	<i>Studie I</i>	Tape
Herbert Eimert	<i>Etüde über Tongemische</i>	Tape

p. 61 Cage and Tudor's Donaueschingen concert was controversial, to say the least. As Joan Peyser writes, "Everywhere Cage and Tudor went, they were treated like a couple of clowns. At Donaueschingen the audience was angry and the press outraged. Complaints focused on the 'noise.'"⁵¹ In another account, the Donaueschingen audience apparently responded to Cage and Tudor's performance "with a mixture of spontaneous pleasure and simultaneous outrage, shouting, laughter, and confusion."⁵²

Cage and Tudor seem to have enjoyed a somewhat smoother reception at the *Musik der Zeit* concert in Cologne a few days later, not least because of the active rivalry between the two West German new music scenes.⁵³ For example, one reviewer rhetorically raised his eyebrows at a "piano with paper strips and paper clips," but focused his blurb on the importance of Cologne being first to host premieres, and ultimately first in the longer musical-historical purview.⁵⁴ In the logic of the cultural Cold War, the West German impresarios in charge of the Donaueschingen and Cologne concert series competed to be on the right side of history.

The honoraria that Cage and Tudor received for the concerts at Donaueschingen and Cologne can further clarify the esteem with which new music impresarios regarded Cage. Strobel and the SWF paid Cage and Tudor 2,000 Deutsche Marks (DM)—\$4,330 in 2017 dollars—a lump sum that Cage and Tudor received in advance and used for their travel expenses.⁵⁵ From the WDR, Cage initially requested an honorarium of 3,000 DM—the equivalent of about \$6,370 in 2017 dollars. Eventually, as Figure 2.3 shows, the WDR paid Cage and Tudor 2,000

DM, the same as the SWF. Beyond honoraria, both festivals also undertook considerable costs to rent and transport large Steinway grand pianos for the concerts.⁵⁶

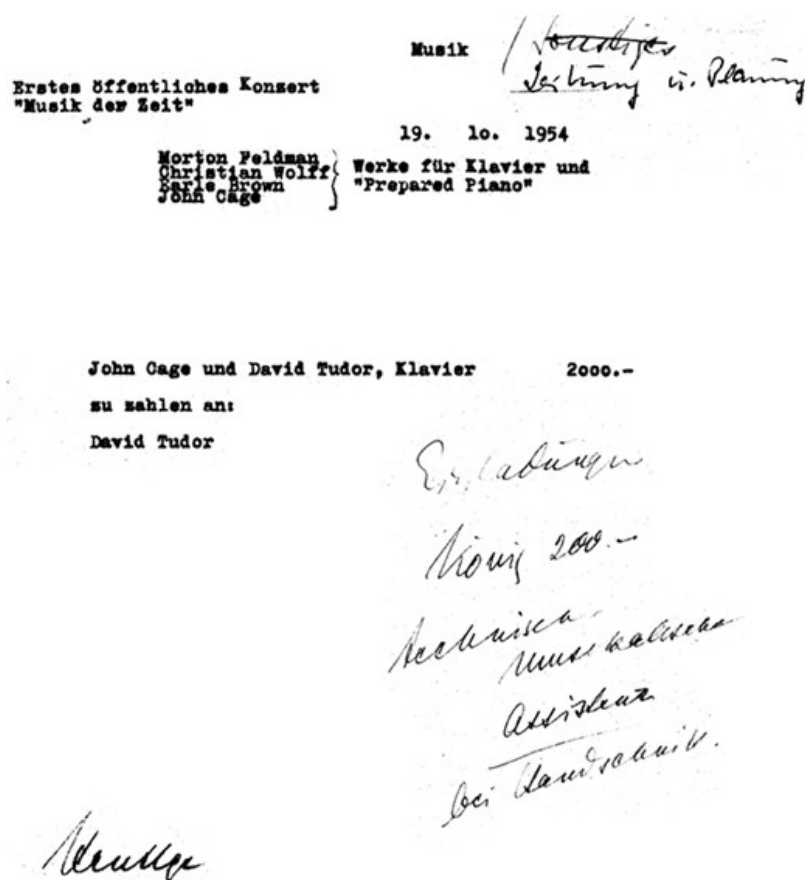


Figure 2.3. Receipt for Cage and Tudor's 1954 *Musik der Zeit* honorarium. WDR Historisches Archiv.

p. 62 Cage and Tudor's substantial honoraria are wildly out of proportion to the stipends that anyone else drew. Goeyvaerts and Pousseur, who each had ↳ their electronic pieces premiered in the second half of the WDR program, were paid 50 DM each, enough to cover their second-class train travel to and from Cologne. The sound engineer made 10 DM per hour; Gottfried Michael Koenig was paid 200 DM for "technical and musical" assistance. Although Cage and Tudor still had to defray international travel costs, they failed to draw such substantial honoraria again at subsequent concerts in West Germany. When they returned to the WDR and *Musik der Zeit* in October 1958 for a performance of the *Concert for Piano and Orchestra* after their visit to Darmstadt, they agreed to an honorarium of 750 DM, at least two and a half times less than what they were

p. 63 paid in 1954. ↳

How might we reconcile the critics' and public's apparent disdain for Cage and Tudor's experimental buffoonery with the outsized honoraria West German administrators paid for their concert appearances? It would seem that in 1954, Cage and Tudor were underground European celebrities. In the new music scene, a number of institutions were competing for funds, notoriety, and international prestige.⁵⁷ Cage and Tudor's *succès de scandale* appearances in Donaueschingen and Cologne fueled the fire; they gained much more from being talked about—even if in the negative—than they did from being passively accepted, or worse, ignored. As the saying goes, there's no such thing as bad press. Strobel and the SWF embraced Cage's notoriety, leveraging his *provocateur* reputation toward the visibility his concert series stood to gain; the same could be said for Eimert and the WDR.

Cage and Tudor were reliable harbingers of the new, and they offered much by way of symbolic capital for image-conscious new music impresarios trying to mark their venue—Darmstadt, Cologne, Donaueschingen, Paris, etc.—as *the* place for European new music. Cage and Tudor made a lasting imprint on those powerful administrators, impresarios, and influential young composers who were largely controlling the new music scene. Those power brokers, in turn, promulgated Cage's ideas and works among the European avant-garde.⁵⁸ As the American experimental composer Earle Brown reflected, the most important introductions Boulez made for him in Europe were to Hans Rosbaud (a conductor of the SWF orchestra who was capable with new music scores), Otto Tomek (an editor at Universal Edition), and William Glock (a British critic, impresario, new music advocate, and controller at the BBC).⁵⁹ The institutional support for new music in West Germany turned on the judgments of a few powerful administrators who had the power to determine broadcast content, concert programs, funding streams, and guest appearances. Ultimately, Cage's embrace in institutional circles in West Germany in 1954 was far more consequential than a few negative reviews in the press.

The impact of Cage's music was further amplified at the WDR electronic studio, where his recordings were archived in the *Handapparat* library, an ad hoc collection of recordings deemed important by Eimert or other composers. The collection included electronic pieces produced in the studio as well as rare recordings of other recent music.⁶⁰ Koenig remembered first encountering Cage's prepared-piano music by means of WDR recordings, perhaps between the 1952 radio broadcast and Cage and Tudor's 1954 visit.⁶¹ Visiting composers such as Mauricio Kagel, György Ligeti, Bengt Hambraeus, and Paul Gredinger commonly began at the WDR studio with a long listening session in the *Handapparat* recordings—Ligeti called this the “best shock of his life.”⁶² Through one or another of these avenues—live concerts, radio broadcasts, or recorded archives—the WDR-affiliated composers stoked their genuine interest in Cage's early music.

Widening Squares

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We have already seen how the sound of Cage's prepared piano provided a timbral model for Eimert's, Schaeffer's, and Stockhausen's early electronic music. As it happens, Cage's early works may have also offered a crucial inspiration when it came to duration. In the electronic studio, composers confronted the new challenge of determining durations and temporal pacing of their music without the conventional rhythmic and metric structure of notation. Instead of quarter-notes, dotted-sixteenth-rests and so on, composers planned durations in clock time, often calculating seconds into corresponding centimeters of tape and vice versa—this is the “rhythmicizing” that Eimert spoke about in Chapter 1. This durational thinking was aided by the materiality of tape: a length of tape corresponded in a consistent and transparent way to the duration of a sound or silence.⁶³ On the one hand, this simple correspondence could be seen as more intuitive than the abstract, symbolic language of rhythmic notation; on the other hand, tape lengths are clumsy. At the standard tape speed of 76.2 cm/s, one can imagine that splicing and montaging durations longer than a second or two could very quickly become unwieldy.

One direct correspondence between Cage's early prepared-piano works and the WDR tape works is the use of durational notation, which Cage used in his “time-length” works (such as 31'57.9864" and 34'46.776"). In these pieces, Cage forgoes the usual system of rhythmic notation, preferring instead an analog that is more like a written-out version of tape music, wherein visual distance corresponds to duration. As Figure 2.4 illustrates, the pitches that the pianist is to play are given on a traditional grand staff, but above, the pianist reads a graphic notation to determine articulation, dynamics, and temporal placement of attacks.

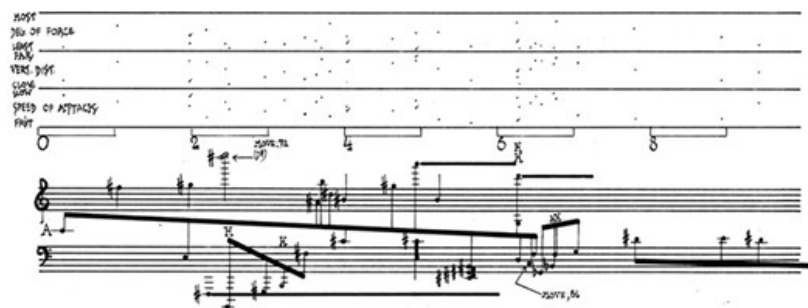


Figure 2.4. Cage, 31'57.9864" for a Pianist, score excerpt from p. 1, showing durational notation.

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Scholars have advanced a number of possible explanations for Cage's evolving durational thought, from his extensive experience playing for dancers, to his increasing immersion in the paradoxes of Zen Buddhist teachings, to observing the way David Tudor used a stopwatch to accurately perform the durations in *Music of Changes*.⁶⁴ These explanations, while plausible, overlook the fact that Cage had been composing *Imaginary Landscape No. 5* and *Williams Mix* in the privately owned studio of Louis and Bebe Barron since early 1952.⁶⁵ It is probable that Cage's durational notation evolved out of his own experiences with composing tape music.

As a point of comparison, consider Stockhausen's *Studie II*, one of the only WDR pieces to include a score that is both a visual rendering and production directions. As Figure 2.5 shows, frequency blocks—additive synthesis mixtures—are shown on the top staff of *Studie II*'s score, tape lengths (in centimeters) on the middle, and decibel curves for dynamic changes on the bottom. Both Cage's time-length works and Stockhausen's electronic tape music use a direct-duration system, whereby the length of tape or visual distance between events on the score correlates directly to the temporal unfolding of the music.

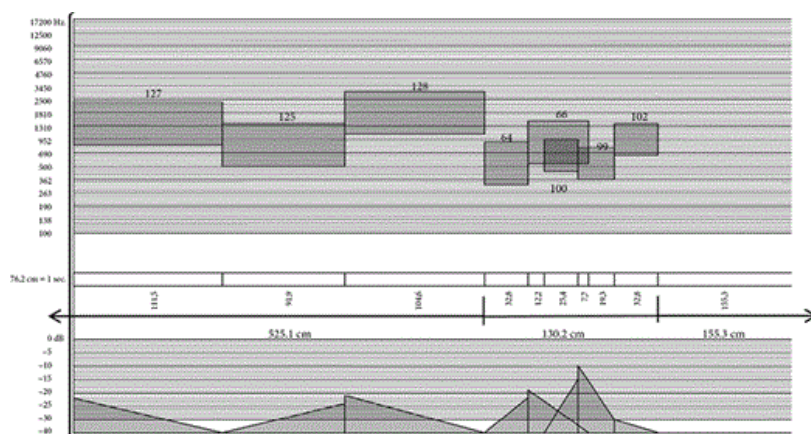


Figure 2.5. Stockhausen, *Studie II*, score excerpt, p. 4. Annotations label mixtures according to the precompositional chart and tape lengths in centimeters; cf. Tables 2.4 and 2.5. Stockhausen Foundation.

The similarities between Cage's and Stockhausen's handling of durations are deeper than the visual correspondences in their scores. In fact, it was duration—not aleatory pitch techniques—that most interested the young European composers, for instance in "Alea," where Boulez praised Cage's durations for inducing a rather undirected, improvisatory (read: Eastern) musical unfolding that is more like an open circle than a (Western) "closed circle of possibilities."⁶⁶ Boulez's momentary concession, a respite in his otherwise scathing critique, continued his long-standing endorsement of Cage's handling of duration. Already in his 1949 introduction at Tèzenas's salon, Boulez had praised Cage's innovative rhythmic structures, which "take into account [. . .] the passing of real time," and result in musical-temporal structures that are either "prismatic" or

“crystallized.”⁶⁷ Boulez seemingly had some understanding of square-root form even then, as he explained that Cage used both whole-number and fractional proportions in creating the temporal structure of his works.⁶⁸ Cage further clarified square-root form to Boulez in a letter in 1950, wherein he analyzed *First Construction (in Metal)*, one of the pieces well known to Boulez, Eimert, Stockhausen, and many others in the European avant-garde.⁶⁹ It is on these grounds that Boulez proposed that he and Cage were linked in “researching a work’s structure by means of rhythmic structures.”⁷⁰

Cage used the proportional technique of square-root form between 1939 and 1956, and *First Construction* is a classic example.⁷¹ The basic premise is this: the proportional scheme 4:3:2:3:4 governs the relationship between elements at all formal levels of the composition, including sections, phrases, and rhythmic motivic cells. These nested proportional relationships are ↵

↵ schematic, but also aurally salient. In *First Construction*, various percussion instruments alternate phrase by phrase, such that different instrumental timbres articulate the changes between phrases of 4-, 3-, and 2-bar lengths. Square-root form structures the piece’s sounds within a tight, iterative, proportional hierarchy.

It appears that Stockhausen strove to replicate the microproportional–macroproportional relationships so familiar from Cage’s square-root form in his first, discarded *musique concrète* piece. As he explained to Goeyvaerts, “I now wanted a structure, to be realized in an *Étude*, which was already worked into the micro-dimension of a single sound, so that in every moment, however small, the overall principle of my idea would be present.”⁷² Cage’s early works were a model because, to paraphrase Stockhausen, Cage’s musical structure was finally based on sounds themselves; the alternation between sounds and silences played a crucial role in the form of the composition.⁷³ According to Stockhausen, Cage’s timbrally and durationally innovative work “heralds the multiplicity of relationships and changing states between material and material forms, whereby the smallest and the largest and the detail within the whole can be deeply bound together.”⁷⁴

As we will see in Chapters 3 and 4, proportions were of the utmost importance for the serial composers, who by 1953 began to focus on proportional relationships between terms in the series rather than using the series simply to order musical events. We will see in Chapter 4 that the kaleidoscopic durational structure of Koenig’s electronic composition *Essay* could easily be related to the microdurational–macrodurational hierarchy on display in Cage’s square-root pieces. Likewise, it should come as no surprise that Stockhausen worked extensively with both frequency and durational proportions in his early electronic music. In Stockhausen’s first WDR piece, *Studie I* (1953), his frequencies were based on a proportional series involving the whole-number integers 4, 5, 8, and 12 (see Figure 3.3 in Chapter 3). Stockhausen continued his work with proportions in *Studie II* (1954). To generate the pitch (frequency) material for the piece, he used an irrational number, the twenty-fifth root of 5—that is, $\sqrt[25]{5}$, or 1.066495—to build a scale with a constant interval.⁷⁵ He then created five-frequency mixtures beginning at 100 Hz, using every scale step (1:1), every other scale step (1:2), and so on to 1:5. This created a proportional relationship whereby subsequent mixture groups were related to the basic unit by the proportions 1:2, 1:3, 1:4, and 1:5.

Stockhausen’s durational structure in *Studie II* was likewise attentive to proportions, as can be seen in an excerpt between 0:27 and 0:34. Table 2.4 shows Stockhausen’s frequency mixtures, their tape lengths in centimeters, their correlated durations in seconds, and their approximate temporal proportions. The end of this passage is shown, with score annotations, in Figure 2.5. Although we do not see the first two mixtures under discussion in Figure 2.5, Table 2.4 clarifies that the five mixtures at the beginning of the ↵ excerpt are quite similar. They are all drawn from the same segment of the precompositional chart (mixtures 125–129), where the mixture frequencies are spaced every four scale steps. Furthermore, their durations are approximately in the proportions 1:1—the tape lengths vary up to 20 cm, but this produces a difference of at most 0.3 s. To my ear, these durations are close enough to be in the same proportion.

Table 2.4. Mixtures, tape lengths, durations, and proportions in Stockhausen’s *Studie II*, 0:27–0:34. For a visual representation of mixtures 127, 125, and 128, refer back to the score excerpt in Figure 2.5.

Mixture	Tape Length (cm)	Duration as Heard (s)	Approx. Proportion
129	98.1	1.287	1
126	119.0	1.561	1
127	111.5	1.463	1
125	91.9	1.206	1
128	104.6	1.372	1

Table 2.5 clarifies the relationship between these five long sounds (mixtures 125–129) and the music that follows. Durationally, the five long sounds together are four times as long as the second event, which is a conglomerate of fast sounds (mixtures 64, 66, 99, 100, and 102). This conglomerate is about 130 cm long (~1.7 s) and exists in approximately equal proportion to the silence of 155 cm (~2 s) that follows it. Figure 2.6 shows this in schematic form. On the foreground level at the beginning of our excerpt, the five similar mixtures effectively have 1:1 proportional duration relationships with each other. At the phrase level, these five long sounds together exist in a 4:1:1 proportion with the fast-sound conglomerate and with the silence that bookends these two gestures. 4

Table 2.5. Mixtures, tape lengths, durations, and proportions in Stockhausen’s *Studie II*, 0:27–0:38. For a visual representation of duration lengths and mixture conglomerates, refer back to the score excerpt in Figure 2.5.

Mixture	Tape Length (cm)	Duration as Heard (s)	Approx. Proportion
Subtotal	525.1	6.891	4
125–129			
Subtotal	130.2	1.709	1
64, 66, 100, 99, 102			
Silence	155.3	2.038	~1

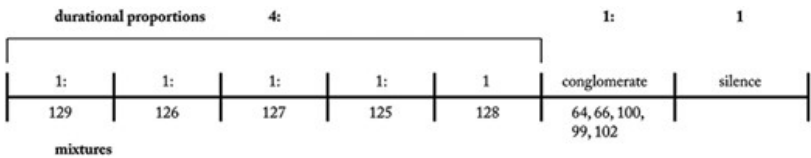


Figure 2.6. Schematic representation of temporal proportions in Stockhausen’s *Studie II*, 0:27–0:38.

This durational proportional structure is approximate, but perceptually functional. To add one further layer of complexity, recall that the five long sounds (mixtures 125–129) are all drawn from the frequency groups with the 1:4 ratios. Perhaps it is not coincidental that the frequency bands with the 1:4 ratio are deployed in a 4:1:1 durational relationship with the following gestures. It is here, in the proportional relationships between

frequency and durations, that we can see Stockhausen manifesting his desire to bind together the smallest and largest detail within the whole work. The durational proportions of the *Studie II* excerpt, expressed as they are on both foreground (mixture) and middle-ground (phrase) levels, closely mirror the micro- and macro-orientations of Cage's square-root form in the *First Construction*. It was Cage's approach that provided Stockhausen with a feasible model for how to enjoin different compositional elements—particularly frequency and duration—into proportional relationships.

Studio Networks

As we have been analyzing the ways John Cage's works were sonic and structural inspirations for electronic music, it is worthwhile to remember that both the French and the West German electronic studios made such cross-pollinations and new relationships possible. Electronic studios—the WDR in particular—were steady anchors of the new music scene, fostering year-round dialogue on new music topics not least because of their continuous funding and guest-composer opportunities. Often, the interpersonal connections made in the studio—especially with collaborators who extended one's own toolkit—made significant and lasting impressions. This was certainly the case for Stockhausen, who, beginning with the 1954 Cologne visit, expressed more interest in David Tudor than in Cage himself.⁷⁶ As he reported to Pousseur,

At the moment, David Tudor is here and is taking up the rest of “free” time. He is stimulating, and the cosmos of imagination of the Cage Group is a monster that is to be taken seriously. [. . .] This is the case ↵ with Tudor (who is speaking for the “Cage Group” at the moment), Brown, Wolff, and Feldmann [*sic*].⁷⁷

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Cage's notoriety may have garnered the pair their first European concert tour, but by the time they arrived in Cologne in 1954, the younger composers were most interested in befriending Tudor, recognizing him to be a prodigious performer and invaluable resource. Stockhausen immediately cultivated an ongoing relationship beginning with a set of *Klavierstücke V–VIII*, which Tudor performed in New York on December 16, 1954.⁷⁸ After this, Tudor went on to perform Stockhausen's *Klavierstück XI* and Boulez's *Troisième Sonate*, as well as works by several other composers who were strongly associated with the WDR studio milieu.

To some degree, these connections were made at the Darmstadt *Ferienkurse*, where Tudor visited four times in quick succession (1956, 1958, 1959, and 1961).⁷⁹ Figure 2.7 shows the repertoire that Tudor played at Darmstadt in the years he visited, organized in a network graph by composer nationality.⁸⁰ The size of each node is proportional to the number of pieces, performances, or both, that Tudor played. Figure 2.7, however, shows only a limited picture of Tudor's international connectedness, as it collates only the repertoire that Tudor played at the Darmstadt *Ferienkurse*.

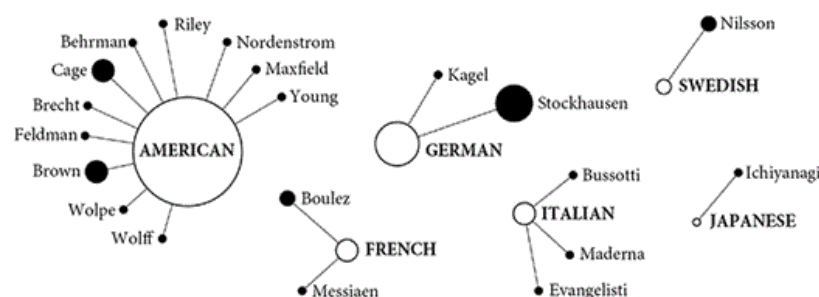


Figure 2.7. Composers whose works Tudor played in Darmstadt between 1956 and 1961, by nationality. Node size is proportional to the frequency with which Tudor played the composer's work.

Figure 2.8 offers a more complete picture of European–American relationships as gleaned from Tudor’s concert programs from 1954 through 1960, with node size again proportional to the number of times composition(s) were played.⁸¹ Predictably, the largest nodes here are the American experimental composers in Tudor’s inner circle, including Cage, Feldman, Brown, and Wolff. We also see that Tudor gave a fair amount of attention to Stefan Wolpe and Lou Harrison, as well as the music of Europeans such as Boulez, Bo Nilsson, Stockhausen, and Pousseur, whom Tudor encountered through WDR networks. Although these more famous composers formed the bulk of Tudor’s repertoire, notice that Tudor played compositions by a huge range ↴ of Europeans, including lesser-known, WDR-affiliated composers such as Franco Evangelisti, Bengt Hambraeus, Mauricio Kagel, and Koenig.

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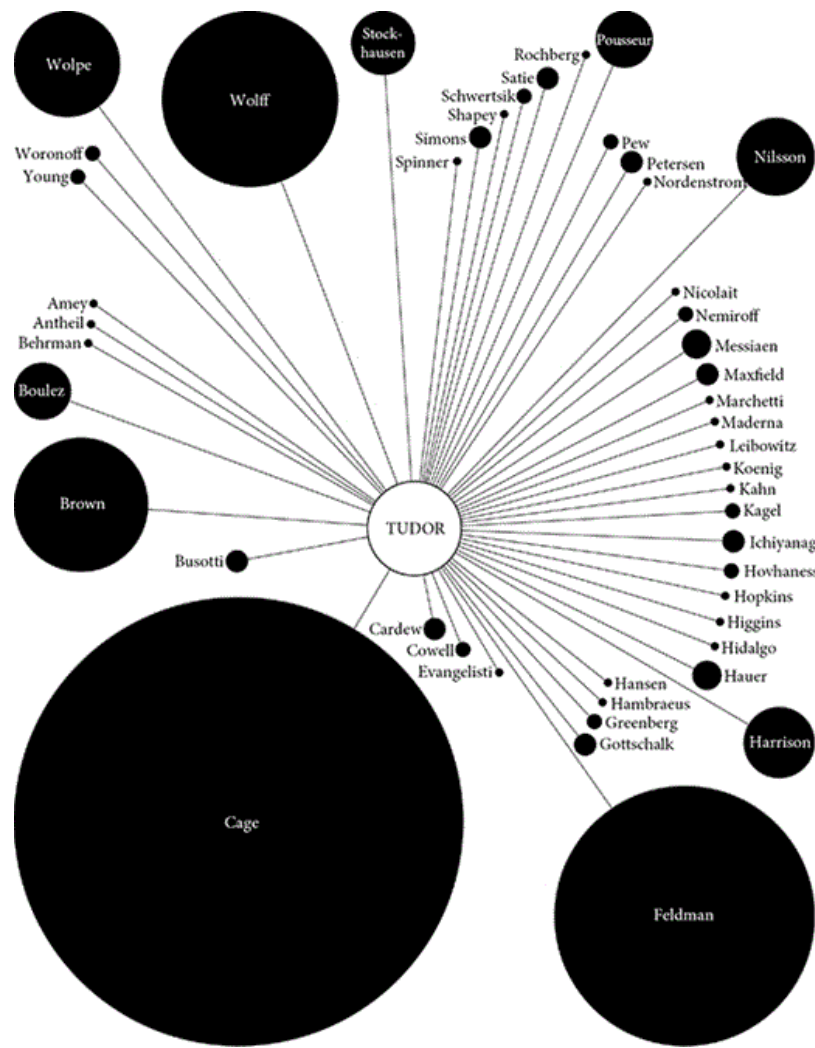


Figure 2.8. Tudor at the center of his international network, based on performances between 1954 and 1960. Node size is proportional to the frequency with which Tudor played the composer’s work.

Network graphs portray Tudor (and the others) as static figures, but in reality, Tudor moved between cities and scenes more than almost anyone else in the European or American avant-garde. From his 1954 Cologne visit on, Tudor was immediately understood as the sole pianist who could perform the technically and conceptually challenging repertoire of both the American ↴ and European avant-garde. This is precisely why Stockhausen, and many others, found that befriending Tudor was so satisfying and efficacious. Tudor’s constant touring, along with his willingness to play new repertoire, afforded a huge range of composers many more good performances, and much more international exposure, than they ever would have gotten without him.⁸²

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Setting Tudor's prodigious talent aside for a moment, we should also remember the institutional mechanisms that allowed such creative partnerships to blossom. The WDR studio offered a physical space for collaboration, where Tudor and the composers first met to study scores, discuss ideas, try out experiments, learn about technological innovations, troubleshoot, and discover solutions to problems. In subsequent years, as Tudor moved between European studios, new music festivals, and his home base of New York City, he functioned as a musical translator, bringing the insights of American and European scenes together in his conversations, programs, and performances. Although Tudor was literally visible on stage and in the studio, his extraordinary contributions remain invisible in a historiographical paradigm that focuses on the innovations of composers.⁸³ In short, Tudor was a central figure in cross-pollinating the international avant-garde.

Conclusions

This chapter has suggested that we nuance our narratives about Cage, Tudor, and the European avant-garde in multiple ways. In summary, Cage and Tudor were relevant—especially to those in electronic music circles—earlier, longer, and far more deeply than is commonly acknowledged. In the early 1950s, the powerful and connected impresarios and composers in the West German and French new music scenes, including Eimert, Strobel, Stockhausen, and Boulez, all embraced Cage. They admired the aesthetic premises of his early music, going so far as to interpret Cage and the American experimentalists as symbols of freedom, democracy, and artistic autonomy. In the cultural Cold War environment of West Germany, impresarios were eager to appropriate Cage's considerable symbolic capital in order to mark their new music scene (Darmstadt, Donaueschingen, or Cologne) as the most progressive.

p. 73 Beyond the prestige that Cage drew, his music was a sonic and structural inspiration for early electronic music both in France and West Germany. Eimert, Schaeffer, and other early advocates of electronic music hailed the prepared piano as a timbral model for the kinds of intriguing new sound worlds they wanted to create. Stockhausen, Boulez, Koenig, and others deeply admired Cage's proportional thinking, particularly with regard to duration. Cage's square-root forms provided a new inspiration for relating duration and pitch proportionally in the studio and for creating multifaceted, nested forms. For nearly a decade before the Darmstadt kerfuffle in 1958, Cage had ↵ been a kindred spirit for the WDR group. In his work with both timbre and proportional durational structure, Cage helped lay the groundwork for electronic music.

Perhaps more lastingly, Tudor was a key figure whose importance to both the European and American avant-garde cannot be overstated. He sustained a robust network that promoted new music compositions and performances, knitting together the new music scenes in locales such as New York City, Paris, Cologne, and Darmstadt. Although Cage continued to be important—if infuriating—to the Europeans in the late 1950s and 1960s (as discussed in Chapter 5), the composers' connection to Tudor remained unsullied. They—Tudor and a large, varied group of avant-garde composers—relied upon each other for creative validation and career advancement.

Multifaceted networks, which include performers, administrators, promoters, impresarios, critics, and composers, can reshape our understanding of the intellectual dimensions, aesthetic concerns, and power dynamics of the Cold War new music scene. With these more expansive networks in mind, controversies such as Darmstadt 1958 become moments in which certain figures become visible as agents, in Latourian terms: they consolidate their power, accrue or spend their symbolic capital, sideline dissidents, recruit collaborators, and so on.⁸⁴ The WDR studio provided one of the most steady, institutional spaces in which such networks could constantly be formed, re-formed, and extended, as new collaborators joined and new issues emerged. In the next chapter, we will see that the WDR studio forged its key identity by perpetuating such a laboratory-like interdisciplinary working method, in which the younger generation of composers formed connections with scientific collaborators in order to advance their aesthetic work. ↵

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Notes

1. Decroupet, "Aleatorik und Indetermination—Die Ferienkurse als Forum der europäischen Cage-Rezeption," in *Im Zenit der Moderne*, vol. 2, 231–40; Griffiths, *Cage*, 30–38; Gann, *There's No Such Thing as Silence*, 193; Shultis, "Cage and Europe," 20–40; Taruskin, *Oxford History of Western Music*, vol. 5, 55, 64–65.
2. The most detailed and nuanced account of the event, which I rely on here, is Iddon, *New Music at Darmstadt*, 196–303.
3. Cage, "Changes" (18–34), "Indeterminacy" (35–40), and "Communication" (41–56), in *Silence*.
4. Iddon, *New Music at Darmstadt*, 216–20.
5. Shultis, "Cage and Europe," 38.
6. Nattiez, "Introduction," *The Boulez-Cage Correspondence*, 3–26; Cage, *The Selected Letters of John Cage*, 103, 111; Haskins, *John Cage*, 51–70.
7. Boulez to Cage, May 7–21, 1951, *The Boulez-Cage Correspondence* #27, 91.
8. Cage to Boulez, May 22, 1951, *The Boulez-Cage Correspondence* #28, 94.
9. See Kostelanetz, *John Cage (ex)plain(ed)*, 7–24; Griffiths, *Cage*, 21–37; Revill, *The Roaring Silence* 107–61; Richards, *John Cage As . . .*, 7–18 and 89–103; Nyman, *Experimental Music*, 50–71; Kahn, "John Cage: Silence and Silencing," 556–98.
10. Cage, "Composition," in *Silence*, 59.
11. Boulez, "Possibly . . .," 112–13.
12. Beal, *New Music, New Allies*; Carroll, *Music and Ideology in Cold War Europe*; Fosler-Lussier, *Music Divided: Bartók's Legacy*; Fosler-Lussier, *Music in America's Cold War Diplomacy*; Shreffler, "'Music Left and Right,'" 67–87.
13. Taruskin, *Oxford History*, vol. 5, 17.
14. Boulez to Cage, December 1951, *The Boulez-Cage Correspondence* #35, 112.
15. Boulez to Cage, July 1954, *The Boulez-Cage Correspondence* #45, 150.
16. Boulez, "Alea," 26–38.
17. Beal, "Negotiating Cultural Allies," 121–23; Beal, "The Army the Airwaves," 487–90; Beal, *New Music, New Allies*, 55–64.
18. Iddon, *New Music at Darmstadt*, 228.
19. Eimert's broadcast transcript from November 27, 1952 is held in WDR Historical Archives, folio 15198.
20. On Cage's early influences, see Bernstein, "John Cage Arnold Schoenberg, and the Musical Idea," 15–45; Joseph, "'A Therapeutic Value for City Dwellers,'" 135–75; and Pritchett, *The Music of John Cage*.
21. Eimert to Stockhausen, September 23, 1952, Stockhausen Foundation, Kürten. "Die Sachen von Cage sind interessant und völlig unbelastet von herkömmlichen Begriffen, darunter eine 'Konstruktion in Metall' und Musik für Prepared Piano, die ich senden werde."
22. Boulez to Cage, October 1, 1952, *The Boulez-Cage Correspondence* #39, 134.
23. Boulez's introduction to Cage's *Sonatas and Interludes*, *The Boulez-Cage Correspondence*, 30.
24. My thanks to David Bernstein who helped me sort out likely corresponding pieces for Eimert's unclear titles and mistaken attributions.
25. Varèse and Wen-chung, "The Liberation of Sound," 11–19.
26. Eimert to Stockhausen, September 23, 1952, Stockhausen Foundation, Kürten. "Die elektronischen Bänder von ihm sind dagegen darg primitiv und geben nichts weiter als ein paar Rückkoppelungstöne."
27. Eimert, *Nachtprogramm* transcript, 7. In the transcript, Eimert also speaks of "eine Musik ohne Psychologie, und die Entwicklung der musikalischen Ideen soll dabei ähnlich vor sich gehen wie die Bildung von Kristallen . . . Und das ist nun wieder einer der jüngsten Begriffe: 'Authentische Musik.'"
28. Eimert, "Was ist elektronische Musik?," 5.
29. Eimert to Stockhausen, March 7, 1952, Stockhausen Foundation, Kürten.
30. Eimert to Stockhausen, March 7–10, 1952, Stockhausen Foundation, Kürten.
31. Eimert to Stockhausen, March 7, 1952, Stockhausen Foundation, Kürten.
32. Eimert to Stockhausen, April 15, 1952, Stockhausen Foundation, Kürten.
33. Schaeffer, *In Search of a Concrete Music*, 18.
34. Goeyvaerts, "Paris: Darmstadt 1947–56," 40.
35. Kurtz, *Stockhausen: A Biography*, 55.
36. Toop, "Stockhausen's Electronic Works," 151.
37. Boulez, "Possibly . . .," 111–40; excerpted in *The Boulez-Cage Correspondence* #37, 128–29. Emphasis in original.
38. Cage, *Bacchanale*. For more on how and with what Cage prepared the piano, see Pritchett, *The Music of John Cage*, 23.
39. Cage, *Bacchanale*. In my discussion, the piano octaves are numbered from the bottom up, such that C4 is middle C.
40. Manning, *Electronic and Computer Music*, 55.
41. Kirchmeyer, *Kleine Monographie über Herbert Eimert*, 48; Manning, *Electronic and Computer Music*, 41.

42. Beal, "The Army the Airwaves," 491.
43. On Strobel's musical and journalistic activities in wartime France, where he was an effective conduit between the Nazi and Vichy regimes, see Fulcher, *Renegotiating French Identity*, 43–44 and 113–14.
44. Cage to Eimert, February 13, 1954, WDR Historical Archive folio 10662.
45. Beal, "The Army the Airwaves," 491–96.
46. Quoted in Beal, "The Army the Airwaves," 493.
47. Cage to Kruttge, April 9, 1954, WDR Historical Archive folio 10662.
48. Cage to Kruttge, May 10, 1954, WDR Historical Archive folio 10662.
49. Kruttge to Cage, June 1, 1954, WDR Historical Archive folio 10662.
50. Beal, "The Army the Airwaves," 494.
51. Peyser, *To Boulez and Beyond*, 178.
52. Beal, "The Army the Airwaves," 497; see also Iddon, *New Music at Darmstadt*, 160–61; Shultis, "Cage and Europe," 31–33; Taruskin, *Oxford History*, vol. 5, 65.
53. Custodis, *Die soziale Isolation der neuen Musik*, 104–108; Hilberg and Vogt, eds., *Musik der Zeit 1951–2001*, 35.
54. E. R. T., "Klavier mit Papierstreifen und Büroklammern," WDR Historical Archive 02240.
55. Beal gives the figure of "several hundred dollars" in "The Army, the Airwaves," 499. She has confirmed in private correspondence that the SWF paid Cage and Tudor 2000 DM, which was the 1954 equivalent of several hundred dollars—about \$476. For the conversion of historical Deutsche Marks to dollars, <http://fxtop.com/en/currency-converter-past.php>; for the conversion of 1954 dollars to 2017 dollars, <http://usinfationcalculator.com> (both accessed September 4, 2017).
56. Beal, "The Army the Airwaves," 493–94.
57. Custodis, *Die soziale Isolation*, 9–27.
58. Beal, "The Army the Airwaves," 497; Custodis, *Die soziale Isolation*, 104–107.
59. Beal, "An Interview with Earle Brown," 341.
60. Digitized recordings of the *Handapparat* library are available at the WDR Historisches Archiv in Cologne.
61. Unpublished portion of an interview between the author and Gottfried Michael Koenig, The Hague, Netherlands, May 10, 2015. It seems likely that the *Handapparat* library contained recordings used on the *Nachtprogramm*, including Cage's *Sonatas and Interludes*, *Bacchanale*, *Three Dances*, and *First Construction*, as well as a recording of the 23' 56.176" and related prepared-piano pieces from the 1954 concert.
62. Ligeti, "Auswirkungen der elektronischen Musik," 86.
63. Kane, "Relays," 66–75; McMurray, "Once Upon a Time," 25–48.
64. The last explanation is favored by Pritchett in *The Music of John Cage*, 102.
65. Pritchett, *The Music of John Cage*, 90–91; Holmes, *Electronic and Experimental Music*, 108.
66. Boulez, "Alea," 30.
67. *The Boulez-Cage Correspondence*, 31.
68. *The Boulez-Cage Correspondence*, 31.
69. Cage to Boulez, January 17, 1950, *The Boulez-Cage Correspondence* #7, 49–50.
70. Boulez to Cage, January 3, 11, 12, 1950, *The Boulez-Cage Correspondence* #6, 44.
71. Square-root form is explained most carefully in Bernstein, "In Order to Thicken the Plot," 7–40; see also Shultis, "No Ear for Music," 83–104; van Emmerik, "An Imaginary Grid," 217–37; Jenkins, "Structure vs. Form in the *Sonatas and Interludes for Prepared Piano*," 239–61; Pritchett, *The Music of John Cage*, 15–22.
72. Stockhausen to Goeyvaerts, December 3, 1952. Translated in Toop, "Stockhausen's Electronic Works," 151.
73. Stockhausen, "Arbeitsbericht 1952/53: Orientierung," 34. Boulez raises similar points in "Possibly . . ." (111–40) and "Tendencies in Recent Music" (173–79) in *Stocktakings*, respectively.
74. Stockhausen, "Arbeitsbericht 1952/53," 34.
75. 25th root of 5 can also be expressed as the fractional exponent $5^{1/25}$.
76. Beal, *New Music, New Allies*, 78–98; Holzaepfel, "Cage and Tudor," 169–85; Holzaepfel, "David Tudor and the *Solo for Piano*," 137–56; Holzaepfel, "David Tudor and the Performance of American Experimental Music 1950–59"; Nakai, "On the Instrumental Natures of David Tudor's Music"; and the special issue (*Composers Inside Electronics: Music After David Tudor*) of *Leonardo Music Journal* 14 (2004): 1–115.
77. Stockhausen to Pousseur, ca. October 18–19, 1954, Pousseur Collection, Paul Sacher Foundation. "Augenblicklich ist David Tudor hier, mit dem der Rest 'freie' Zeit verbracht wird. Er ist anregend, und die Vorstellungswelt der Cage-Gruppe ist ein ernst zu nehmendes Monstrum. [. . .] Sie haben nichts zu verantworten, nichts zu beraten, keine Tradition. Das ist unheimlich, wenn man es so unmittelbar erkennt, betont, der Beleg benutzt antreffen kann, wie bei Tudor, der in Augenblick (in Namen der 'Cage-Gruppe' spricht) Brown, Wolff, Feldmann."
78. Stockhausen to Pousseur, December 5, 1954, Pousseur Collection, Paul Sacher Foundation. By his own account, Stockhausen gave the last *Klavierstück* manuscript (VI?) to Tudor on December 8 for the performance on December 16—an

extraordinarily short lead time considering international mail. Tudor discusses this a bit in Holzaepfel, “Interview with David Tudor,” 630.

79. Evidently, for administrators at Darmstadt, Tudor was the more desirable half of the Cage–Tudor pair from the start. See Iddon, *New Music at Darmstadt*, 196.
80. Data compiled from Borio and Danuser, eds., *Im Zenit der Moderne*, vol. 3, 577–613.
81. Data compiled from Holzaepfel, *David Tudor and the Performance*, Appendix D, 328–62.
82. Beal, “David Tudor in Darmstadt,” 77–88; Beal, “An Interview with Earle Brown,” 355; Nakai, “On the Instrumental Natures of David Tudor’s Music”; Yaffé, “An Interview With Composer Earle Brown,” 296.
83. Meehan makes a similar argument about Cathy Berberian in “Not Just a Pretty Voice.”
84. Latour, *Science in Action*; Piekut, *Experimentalism Otherwise*; Piekut, “Indeterminacy, Free Improvisation, and the Mixed Avant-Garde,” 769–824.

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